

INSS 2026 Project

Beyond the Standard Model

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1 Generator in a Bottle

Walking along the Isla Vista beach, you find a mysterious bottle with a note and a floppy disk inside. The note reads:

Dear friend,

After many years of isolation and research, I have finally devised the perfect neutrino interaction generator. I was able to solve QCD and electroweak interactions exactly all the way from neutrino energies of 1 meV up to 100 TeV. The only limitations are the values of the fundamental coupling constants, which I have fixed to the best known values today.

My generator takes in a neutrino flux model and a detector geometry file and includes all Standard Model effects to simulate interactions with any target material, including reinteractions of secondary particles.

I have also included a new physics module that allows you to simulate interactions with any new physics model you can think of by simply typing in the Lagrangian of your model.

Use it wisely.

Yours truly,
Dr. X. Boson

With such tremendous power at your fingertips, your team becomes a leader in neutrino physics and obtains unlimited resources to carry out your research.

What experiment would you build and how would you use this mysterious generator to discover new physics?

Some questions to consider:

- What is the physics case for your experiment? What new physics model are you trying to probe?
- What kind of neutrino energy, beam, and detector would you use?
- How could you use your generator to bypass the remaining uncertainties?
- What are the main backgrounds to your signal and how would you mitigate them?
- Can you derive the sensitivity of your experiment to a specific new physics model? What couplings strengths and mass ranges can you probe?

2 Late to the Interaction

While talking to your colleagues in downtown Isla Vista, you hear about a mysterious set of anomalous events that have been observed at the SBND experiment after a single year of data taking. These events are rare, but are observed to appear in the detector later than neutrinos would. Your team decides to investigate this and you are given full access to the SBND data.

- What kind of new physics could explain the time delay on these events?
- What would you look for in the data to understand the nature of these events?
- Depending on what you find, give more details on the physics model and estimate couplings, masses, lifetimes, or cross sections required to explain the observed number of events in SBND.
- How would you prove that your new physics model is the correct explanation for these events?
- What other experiments could you use to confirm your findings?

Feel free to assume whatever properties for the new events makes most sense for your team. You can also tell the story of how you can go from these anomalous events to an eventual discovery of the underlying new physics with future detectors. What are the consequences of your discovery for the field of particle physics? Can you think of any applications for your new physics model in other areas of science or technology?